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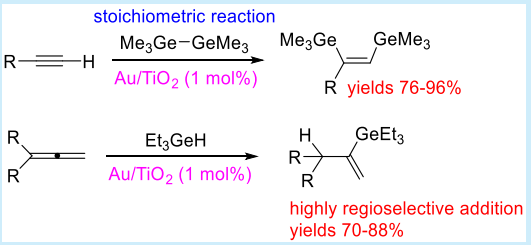
Synthesis of Vinylgermanes via the Au/TiO_2-Catalyzed *cis*-1,2-Digermylation of Alkynes and the Regioselective Hydrogermylation of Allenes

Anastasia Louka and Manolis Stratakis*

Cite This: *Org. Lett.* 2021, 23, 3599−3603 Read Online

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ABSTRACT: In the presence of Au/TiO_2 (1 mol %), terminal alkynes react quantitatively with stoichiometric amounts of the unactivated digermane Me_3Ge-$GeMe_3$, forming exclusively *cis*-1,2-digermylated alkenes. We also establish the Au/TiO_2-catalyzed hydrogermylation of terminal allenes with Et_3GeH, which exhibits a highly regioselective mode of addition on the more substituted double bond forming vinylgermanes. Additionally, we provide preliminary results regarding the Pd nanoparticle-catalyzed C−C coupling of 1,2-digermyl alkenes with aryl iodides.

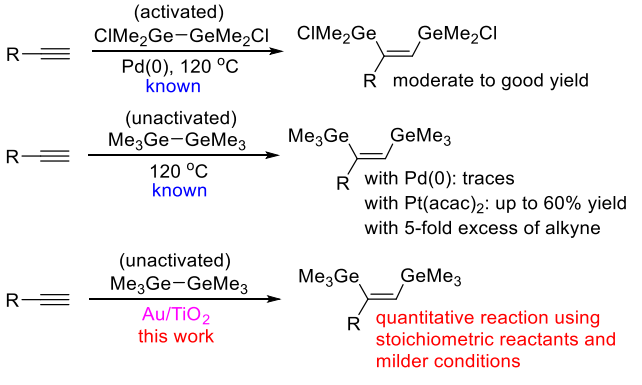


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In contrast to organosilicon compounds whose synthesis and C−C cross coupling reactions have been extensively studied,[1] organogermanes have not received significant attention.[2] Very recently, however, the field of organogermanium chemistry has been revived, particularly by the findings by the group of Schoenebeck, who have uncovered a series of novel and selective metal-catalyzed C−C cross coupling protocols of C(sp^2)-germyl linkages.[3] The main innovation in these cross coupling protocols is that simple trialkylgermyl substituents are being used,[4] in contrast to the previously known cases where the germyl substituents should be activated either by internal coordination/chelation forming hypervalent species, or if bearing labile heteroatom ligands and Si−Ge linkages.[5] As the C(sp^2)-GeR_3 functionality (R = alkyl) appears to be very promising in a palette of synthetic applications, there is need for the development of efficient protocols regarding the selective preparation of vinylgermanes.[6] So far the majority of the existing protocols deal with the hydrogermylation of alkynes.

Our group has in recent years a continuous interest in studying the Au nanoparticle (Au NP)-catalyzed reactions of σ heteroelement linkages (Si−Si, B−B, B−Si) with π systems forming alkenyl silanes or boranes.[7] Thus, in the presence of Au/TiO_2, alkynes react smoothly with 1,2-disilanes (directly or indirectly),[8] silylboranes,[9] or 1,2-diboranes,[10] forming the corresponding *cis*-1,2-adducts. Urged by the recent awakening of organogermanium chemistry,[3] we examined the possible digermylation of alkynes as a practical route for the synthesis of vinylgermanes using Au NPs as the catalyst. The 1,2-digermylation of alkynes is a transformation that has not been extensively studied in the past, and there are only a few synthetically useful protocols in the literature.[11] In the first reported example, a highly strained 3-membered ring digermirane was used, which undergoes insertion on Pd(0) that subsequently adds to alkynes.[12] A few other reports regarding the Pd- or Pt-catalyzed reaction of the rather exotic cyclic oligogermanes with alkynes are also known.[13] As per linear 1,2-digermanes (Scheme 1), the Pd-catalyzed (5 mol %) digermylation of excess of phenylacetylene with the activated $ClMe_2GeGeMe_2Cl$ yields at 120 °C the *cis*-adducts in moderate to good yields, while under the same conditions, the unactivated hexamethyldigermane ($Me_3GeGeMe_3$) provides trace amounts of products.[14] The conversion yield of the digermylation with $Me_3GeGeMe_3$ can increase to 60% using

Scheme 1. Catalytic 1,2-Digermylation of Alkynes



Received: March 23, 2021
Published: April 9, 2021



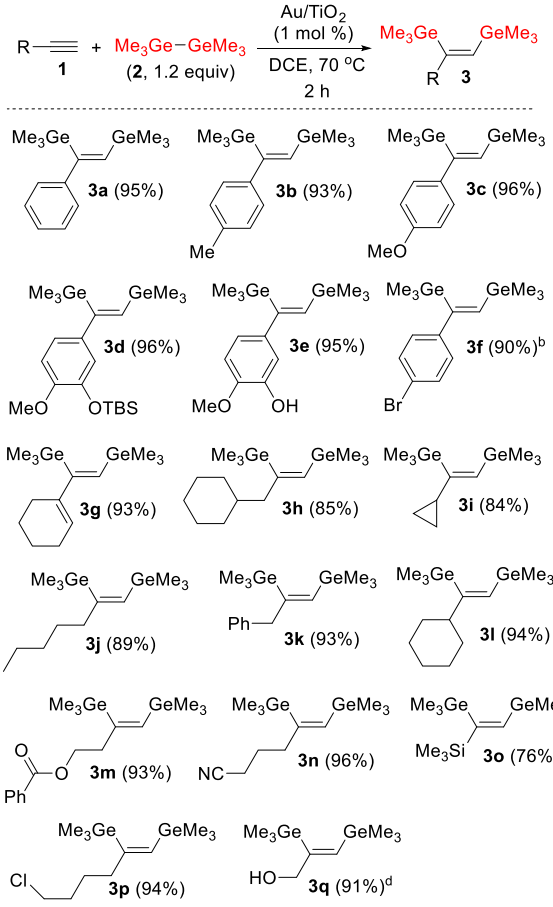
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https://doi.org/10.1021/acs.orglett.1c00997
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$Pt(acac)_2$ as catalyst (5 mol %) at 120 °C but with a 5-fold molar excess of alkyne.[13a]

To our delight, in the presence of Au/TiO_2 (1 mol % in Au), the reaction of a series of terminal alkynes with $Me_3GeGeMe_3$ (**2**, 1.2 equiv) proceeds smoothly in dry 1,2-dichloroethane as the optimum solvent (70 °C), yielding after 2 h the 1,2-*cis*-digermylated adducts in very high isolated yields (Table 1).

Table 1. *cis*-1,2-Digermylation of Terminal Alkynes with $Me_3GeGeMe_3$ Catalyzed by Au/TiO_2[a]



[a]Isolated yields. [b]Two equiv of $(Me_3Ge)_2$, 4 h. [c]Z/E = 75/25. [d]With 2.5 equiv of $(Me_3Ge)_2$, a mixture of **3q** and its *O*-trimethylgermyl protected derivative **3q′** is formed in ∼1/1 ratio (see Supporting Information).

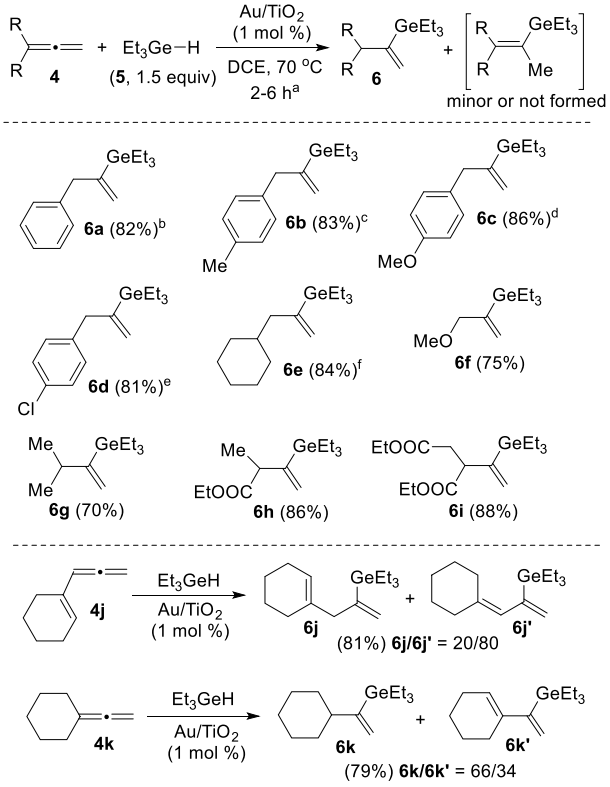
Au/Al_2O_3 and Au/ZnO exhibited lower efficiency (page S3 of the Supporting Information). In the majority of the cases, there was no essential need to perform any chromatographic purification of the adducts, indicative that no side products were formed. The *syn*-1,2-addition of the digermane was confirmed by 2D NMR experiments. The reaction is stoichiometric, as in many cases, the same outcome may be observed using strictly 1.0 equiv of digermane. Given, however, that $Me_3GeGeMe_3$ suffers from partial destruction in the presence of traces of moisture, we used a 20% molar excess in all cases to secure the completion of the digermylation. Notably, the efficiency of our catalytic system is still substantial at 0.1 mol % loading of Au/TiO_2, obtaining ∼40−70% lower yields within 2 h (page S3 of the Supporting Information). Among the examples of Table 1, for reasons that are not fully understood, *p*-bromophenylacetylene (**1f**) reacts slowly, and a 2-fold excess of hexamethyldigermane is required for the reaction to go to completion within 4 h. Also, with stoichiometric amounts of $(Me_3Ge)_2$, propargyl alcohol (**1q**) provides the digermyl adduct **3q**, but with 2.0−2.5 equiv, the trimethylgermyl protected alcohol **3q** (compound **3q′**, see Supporting Information) is also formed in ∼50% yield, indicative of a Au/TiO_2-catalyzed *O*-germylation.[15] Given that in the presence of a Pt catalyst alkynes react poorly with hexamethyldigermane and under rather forcing conditions[13a] and are almost unreactive in the presence of Pd catalysts,[14] our catalytic system is superior as much lower catalyst loading is required, the reaction conditions are milder, the reaction time is relatively short, the reactants are almost stoichiometric, and finally, the isolated yields are excellent. Unfortunately, internal alkynes appear to be completely unreactive, as has also been observed in their attempted Au/TiO_2-catalyzed 1,2-disilylation.[8a] While Au(I) can formally catalyze the addition of digermane to alkynes via an oxidative addition/reductive elimination catalytic cycle,[16] in our hands, a series of Au(I) substances were inefficient catalysts (page S3, Supporting Information) with some reactivity (5% yield) observed in the presence of 5 mol % of $Ph_3PAuNTf_2$ at 70 °C after 24 h. Perhaps this transformation can be optimized after extensive screening of various Au(I)-based catalysts.

Following the very simple catalytic digermylation protocol of alkynes, we turned our attention to the hydrogermylation of allenes as a new method for the synthesis of vinylgermanes using the readily available Et_3GeH. Our group has shown some years ago that the analogous Au NP-catalyzed hydrosilylation of allenes is regioselective, with predominant addition on the more substituted double bond,[17] while with dihydrosilanes the main pathway is dehydrogenative disilylation on the terminal double bond.[18] In the literature, there is only one example dealing with the hydrogermylation of allenes.[19] In this specific report, Ph_3GeH had been used under catalysis either by Pd(0) or Et_3B (radical conditions). The selectivity is peculiar, providing allylic germanes in moderate yields. After optimization of reactants, stoichiometry, and catalyst support (see Supporting Information), we found that the reaction between Et_3GeH (**5**, 1.5 equiv) and a series of terminal allenes (Table 2) occurs readily in DCE within 2−6 h at 70 °C in the presence of Au/TiO_2 (1 mol %) and yields vinylgermanes. For the case of monosubstituted allenes the reaction is faster, and the addition occurs predominately on the internal double bond in 92−99% selectivity. For 1,1-disubstituted which require 6 h to react, the selectivity is >99%. Allenene **4j** provided an interesting result. The anticipated hydrogermylation product **6j** was formed as a minor one, while the major was the conjugated **6j′**. Product **6j′** does not arise from isomerization of **6j** under the reaction conditions. Additionally, vinylidenecyclohexane (**4k**) yielded apart from the anticipated **6k**, the dehydrogenated product **6k′** (in 34% relative ratio), a product selectivity similar to what has been observed in its analogous Au/TiO_2-catalyzed hydrosilylation.[14] The product distributions from **4j** and **4k** are analyzed in the accompanying mechanistic discussion.

From the mechanistic point of view, we consider in Scheme 2 analogous intermediates to those that have been invoked in the corresponding Au/TiO_2-catalyzed disilylation of alkynes[8a] or hydrosilylation[17] of allenes. In the first case, after insertion of the Ge−Ge linkage of digermane on the Au cluster (Au_n) occurs coordination of the alkyne on the $Me_3Ge-Au_n-GeMe_3$ species, followed by addition of the trimethylgermyl moieties

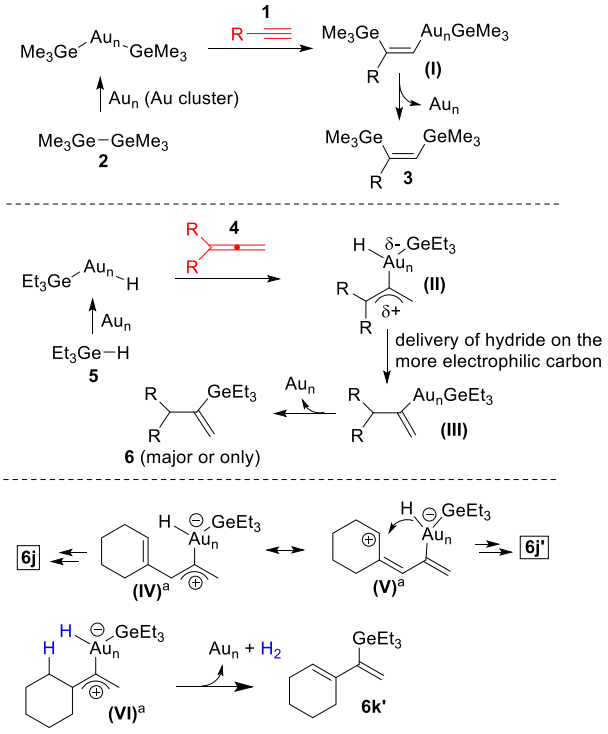
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Table 2. Regioselective Hydrogermylation of Allenes Catalyzed by Au/TiO_2[a]



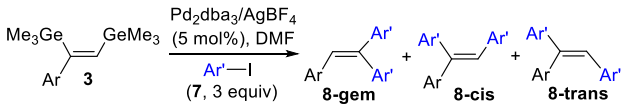
[a]For monosubstituted allenes, the reaction time is 2 h, while for 1,1-disubstituted allenes, it is 6 h. Internal/terminal double bond addition: [b]94/6; [c]92/8; [d]98/2; [e]96/4; [f]99/1.

Scheme 2. Proposed Mechanisms in the Au/TiO_2-Catalyzed *cis*-1,2-Digermylation of Alkynes and Hydrogermylation of Allenes



[a]For intermediates **IV**−**VI**, full charges are shown for clarity.

Scheme 3. Pd-Catalyzed Cross Coupling of *cis*-1,2-Digermylated Adducts with Aryl Iodides



Ar = Ar' = *p*-MePh (**8a**, 72% yield)
Ar = Ph, Ar' = *p*-MePh (**8b-gem/8b-cis/8b-trans** = 55/11/34, 75% yield)
Ar = *p*-MeOPh, Ar' = *p*-FPh (**8c-gem/8c-cis/8c-trans** = 53/23/24, 72% yield)
Ar = Ph, Ar' = *p*-FPh (**8d-gem/8d-cis/8d-trans** = 54/15/31, 73% yield)

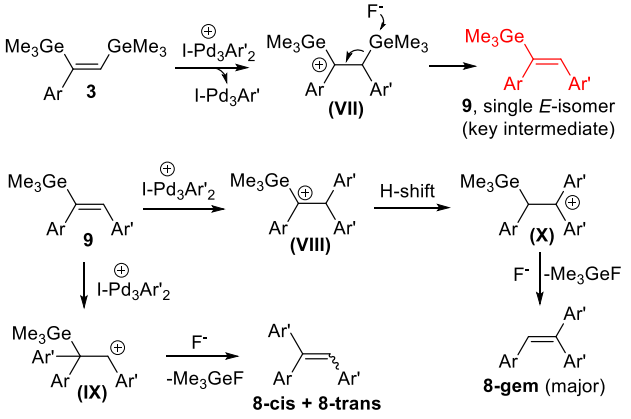
on the triple bond via an inner sphere mechanism (intermediate **I**). Then, the *cis*-1,2-digermylated product **3** is released from the surface of the nanoparticle. Intermediate **I** for the case of the trimethylsilylacetylene is more likely having a dipolar character, as also suggested in the Au/TiO_2-catalyzed disilylation[8a] and diborylation[10] of silylacetylenes, providing a mixture of *E*/*Z* isomers (products **3o*E*/*Z***). In the hydrogermylation of allenes, we consider an η-1-like complex **II** as intermediate, similar to what has been suggested in the addition of X-Y heteroelement linkages to allenes,[17,18,20] which delivers the hydride on the more electrophilic site (more substituted C) to form **III**. This mechanism is consonant to the hydrogermylation of conjugated allene **4j**, for which the anticipated product **6j** is a minor one. Through charge delocalization (structures **IV** and **V** for which fully developed charges are shown for clarity), hydride transfer occurs on the former sp^2-CH of the cyclohexyl ring, leading to the major germyl-bearing conjugated diene **6j′**. The type **II** intermediate complex also rationalizes the formation of the side product **6k′** in the hydrogermylation of vinylidenecyclohexane (**4k**). There, intermediate **VI** may undergo intramolecular elimination of H_2 via a six-membered transition state (bottom part of Scheme 2).

To examine the synthetic utility of the 1,2-digermylated adducts (**3**), as a preliminary study, we attempted their Pd-catalyzed C−C cross-coupling reactions with aryl iodides (Scheme 3) based on the protocol that has recently been developed by the group of Schoenebeck[4a] for the case of aryl triethylgermanes (Pd_2dba_3/$AgBF_4$). The first reaction which was examined was that of the digermylated adduct **3b** with 3 equiv of *p*-iodotoluene. To our delight, the diarylated cross coupling product **8a** (tri*p*-tolylethylene) was formed in 72% yield. However, with different aryl combinations (aryl group in the starting digermylated compound and aryl iodide), a mixture of three isomeric products were seen in the cases that were examined. The major product is diarylated **8-gem** in a relative yield of ∼55%, accompanied by a mixture of **8-cis** and **8-trans** (Scheme 3). In terms of conventional cross coupling aspects of the C−Ge functionalities, the major **8-gem**, which has two newly introduced aryl substituents on the less substituted sp^2-carbon atom of digermyl precursor, is an unexpected one. Despite this interesting observation, the Pd-catalyzed coupling protocol does not appear to be synthetically useful for these substrates. Yet, the accompanying mechanistic analysis for the formation of the three possible isomers is quite interesting. Using 1.5 equiv of aryl iodides instead of 3.0, it was possible to isolate the intermediate monoarylated cross coupling products that were seen in approximately 25−30% relative yield in the crude reaction mixture. Thus, in the reaction between **3b** and *p*-iodotoluene, the only appearing monoarylated germyl product was characterized based on the

https://doi.org/10.1021/acs.orglett.1c00997
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combination of NOESY and HMBC experiments as compound **9a** having an *E*-configuration (Scheme 4 and Supporting Information).[21] Similarly, in the reaction between **3c** and *p*-fluoroiodobenzene, the only monoarylated intermediate **9c** was also isolated and characterized as a single *E*-isomer. Treatment of **9c** with $Pd_2dba_3/AgBF_4$ and 1.5 equiv of *p*-fluoroiodobenzene gave the three diarylated coupling products **8c-gem**/**8c-cis**/**8c-trans** in a relative ratio identical to what is reported in Scheme 3. This result tells us that the *E*-monogermylated **9c** is the only intermediate of the diarylation.

Scheme 4. Proposed Mechanism in the Pd-Catalyzed Diarylation of *cis*-1,2-Digermylated Alkenes with Aryl Iodides



In Scheme 4, we provide some mechanistic clues regarding the product distribution of the three diarylated coupling products, which is based on the mechanism proposed in the Pd-catalyzed Schoenebeck's arylgermane/aryl iodide cross coupling reaction. The *in situ* formed electrophilic Pd_n nanoparticles (we consider herein a hypothetical Pd_3 cluster[4a]) undergo oxidative addition by aryl iodides, and the resulting electrophilic arylated cationic species $I-Pd_3Ar_2^+$ react with digermylated alkene forming intermediate **VII**, which via fluoride ion capture of the trimethylgermyl group yields monoarylated germyl alkene **9** as a single *E*-geometrical isomer. The addition of the electrophilic aryl group occurs exclusively on the less substituted alkene carbon of **3** because it leads to the more stable aryl-stabilized carbocation. Intermediate **9** may subsequently undergo two possible modes of a second aryl addition from the electrophilic aryl-bearing cationic Pd species with predominant C−C bond formation on its less substituted alkene C atom to form **VIII**. Through H-shift on **VIII**, intermediate **X** is derived which, after Me_3GeF elimination, leads to the major diarylated product **8-gem**. We propose that intermediate **VIII** is more stable compared to its alternative **IX** due to the extra stabilization of the carbocation by the germyl group.[22] Intermediate **IX** can lead to the *E*/*Z* isomeric products **8-cis** and **8-trans**. Despite the moderate selectivity in those attempted cross coupling experiments, very useful mechanistic conclusions were drawn that may help in the development of novel cross coupling reactions.

In conclusion, we report herein the so far unrecognized ability of supported Au(0) nanoparticles on TiO_2 in catalyzing the smooth *cis*-1,2-digermylation of terminal alkynes with the unactivated digermane $(Me_3Ge)_2$ and the regioselective hydrogermylation of terminal allenes on their internal double bond, leading to vinylgermanes. Vinylgermanes are compounds which may find potent and novel applications in organic synthesis, and there are limited examples of stereoselective and regioselective methods for their preparation. Our work contributes significantly toward this direction.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c00997.

Copies of ^1H, ^{13}C NMR, and MS spectra of all products (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Manolis Stratakis − *Department of Chemistry, University of Crete, 71003 Heraklion, Greece;* orcid.org/0000-0002-0962-8297; Email: stratakis@uoc.gr

Author

Anastasia Louka − *Department of Chemistry, University of Crete, 71003 Heraklion, Greece*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.1c00997

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The research work was supported by the Hellenic Foundation for Research and Innovation (HFRI) and the General Secretariat for Research and Technology (GSRT) under the HFRI PhD Fellowship grant GA no. 31448.

■ DEDICATION

This paper is dedicated to the memory of Professor Zvi Rappoport (The Hebrew University of Jerusalem).

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